

Step 6 Validation Protocol: Dynamic Phonon Parameter Scans

1 Purpose of the test

Steps 4 and 5 showed that the lattice-dynamic channel follows the imposed spin-Peierls trend. Step 6 isolates the dynamic phonon coordinate itself. The goal is to identify which finite-window observables scale with the operator term $g_Q Q L_{BD}$ rather than with scalar bright-dark mixing.

2 Physical model

The Hamiltonian is

$$H_{\text{mix}} = V_{BD}^{\text{static}} L_{BD} + g_Q Q L_{BD}, \quad (1)$$

with

$$Q = a + a^\dagger. \quad (2)$$

The static component V_{BD}^{static} is held fixed at the low-temperature reference value. The scan therefore changes only the dynamic phonon parameters:

$$g_Q \quad \text{or} \quad \omega_Q. \quad (3)$$

3 Expected results

The $g_Q = 0$ case is the static scalar reference. Increasing g_Q should change at least one finite-window observable relative to that reference: cross-window area, cross/diagonal ratio, sideband/main-cross ratio, full spectrum norm, or pathway-resolved amplitude. For sufficiently weak coupling, changes in intensities are expected to be approximately even in g_Q because the spectrum is insensitive to the sign of a single harmonic displacement in many aggregate observables.

The ω_Q scan should affect windows displaced by $\pm\omega_Q$. If the dynamic sideband interpretation is useful, the sideband-sensitive metrics should vary with ω_Q even when V_{BD}^{static} and g_Q are held fixed.

4 Numerical procedure

Run one parameter set at a time using `STEP6_ACTIVE_CASE`. The recommended cases are:

Scan	Values
g_Q	0, 0.0025, 0.005, 0.010, 0.015, 0.020
ω_Q	0.020, 0.035, 0.050, 0.070 eV

The pulse sequence and NQ protocol are the same as in Step 5: third-order 1Q rephasing spectra with the standard pathway set. The frequency grid is wide enough to include the bright/dark resonances and nearby phonon-displaced windows.

5 Acceptance criteria

The g_Q scan passes if the $g_Q = 0$ static reference is distinct from at least one nonzero- g_Q spectrum in a finite-window metric or full-spectrum norm, with V_{BD}^{static} unchanged. It is inconclusive if all nonzero values look identical to the static reference within numerical tolerance.

The ω_Q scan passes if sideband-sensitive metrics vary with ω_Q . It is inconclusive if the metric is dominated by tails from the main resonances or by finite-window placement rather than by the phonon energy.

6 Requested figures and data

For each run, save:

- the rephasing real spectrum PDF;
- pathway-resolved complex response arrays;
- total rephasing, unrephasing, and 1Q components;
- text and JSON summaries containing V_{BD}^{static} , g_Q , ω_Q , and the finite-grid settings.

Phase 2 should produce scan tables, trend plots for g_Q and ω_Q , and a validation outcome file.

7 Interpretation of possible failures

If the g_Q scan produces no clear change, the coupling may be too weak, the phonon basis may be too small, or the chosen finite windows may be insensitive to the dynamic coordinate. If the ω_Q scan does not move or change the sideband metrics, the observable may be dominated by the main resonances rather than by genuine phonon-displaced pathways.

8 Numerical results

The Step 6 scan generated ten spectra at $\eta = 0.001$ on a 120×120 frequency grid. Throughout this test, $V_{BD}^{\text{static}} = 0.02$ was held fixed, so the observed changes are not caused by scalar bright-dark mixing. The finite-window half-width was 0.025 eV.

g_Q	A_{cross}	$A_{\text{cross}}/A_{\text{diag}}$	$A_{\text{side}}/A_{\text{cross}}$	$\ S_R\ _2$
0	16.266623	0.127003	2.329115	1196548
0.0025	15.927155	0.127904	2.344188	1144065
0.0050	14.997807	0.130900	2.389604	1000153
0.0100	12.554241	0.145423	2.549178	622490
0.0150	11.978064	0.170810	2.468308	442198
0.0200	18.471655	0.201674	1.982344	776222

The g_Q scan clearly changes the nonlinear response even though the static mixing is fixed. The full rephasing norm changes by relative amount 6.30×10^{-1} compared with the $g_Q = 0$ static reference. The cross/diagonal ratio increases monotonically from 0.127003 to 0.201674. The main cross-window area and full norm are not monotonic: both decrease up to intermediate coupling

and then increase at $g_Q = 0.020$. This nonmonotonicity is not a failure; it indicates that the dynamic phonon coordinate changes the spectral redistribution rather than acting as a simple scalar amplitude.

ω_Q	A_{cross}	$A_{\text{cross}}/A_{\text{diag}}$	$A_{\text{side}}/A_{\text{cross}}$	$\ S_R\ _2$
0.020	13.137004	0.134156	3.378482	803047
0.035	12.554241	0.145423	2.549178	622490
0.050	11.581389	0.155038	2.266216	478460
0.070	11.157339	0.160376	3.141335	441798

At fixed $g_Q = 0.010$, increasing ω_Q reduces both the main cross-window area and the full rephasing norm, while the cross/diagonal ratio increases. The sideband/main-cross ratio varies with a relative spread of 3.29×10^{-1} , passing through a minimum near $\omega_Q = 0.050$ eV. This metric is sensitive to ω_Q , but it should be interpreted qualitatively because finite sideband windows can include tails from nearby resonances.

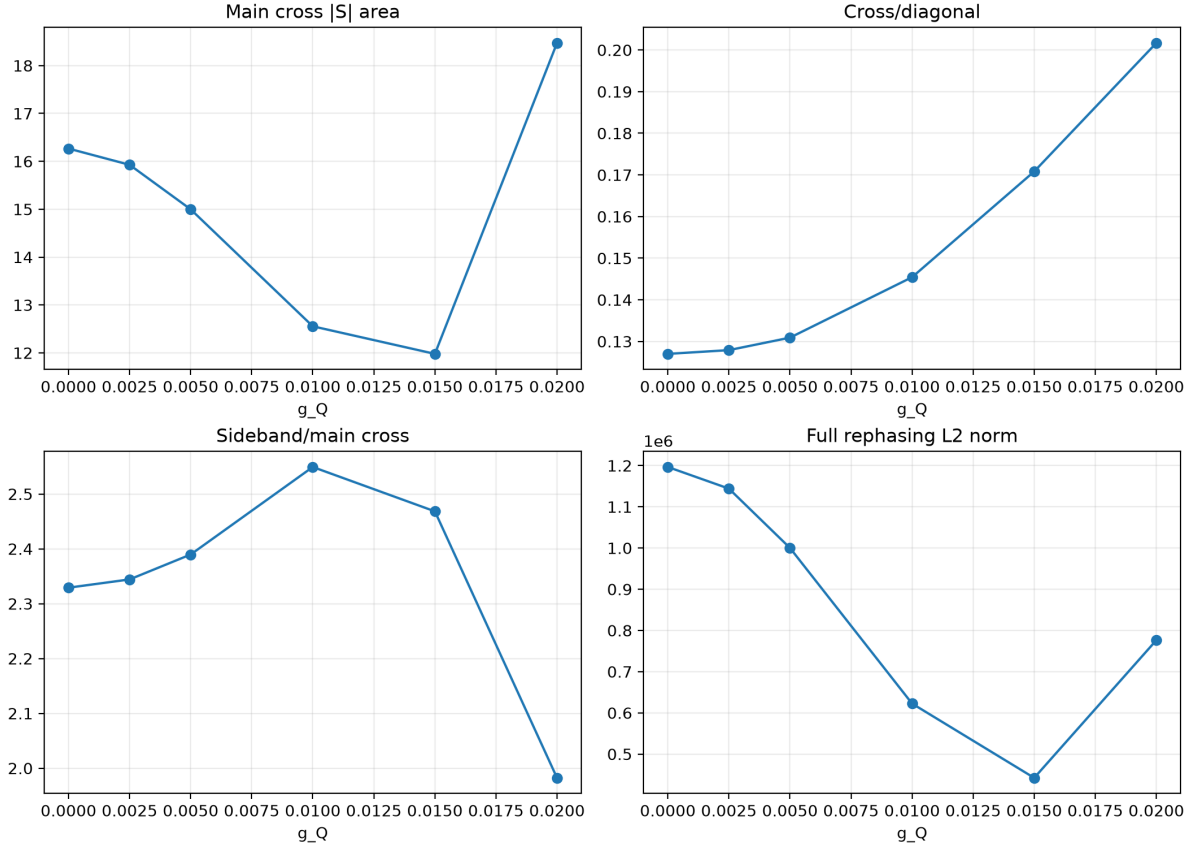


Figure 1: Scan of g_Q at fixed $V_{BD}^{\text{static}} = 0.02$ and $\omega_Q = 0.035$ eV.

9 Validation outcome

The g_Q criterion passes. Nonzero g_Q produces a clear change relative to the static $g_Q = 0$ reference, with fixed V_{BD}^{static} . Therefore the spectral change cannot be attributed to scalar bright-dark mixing

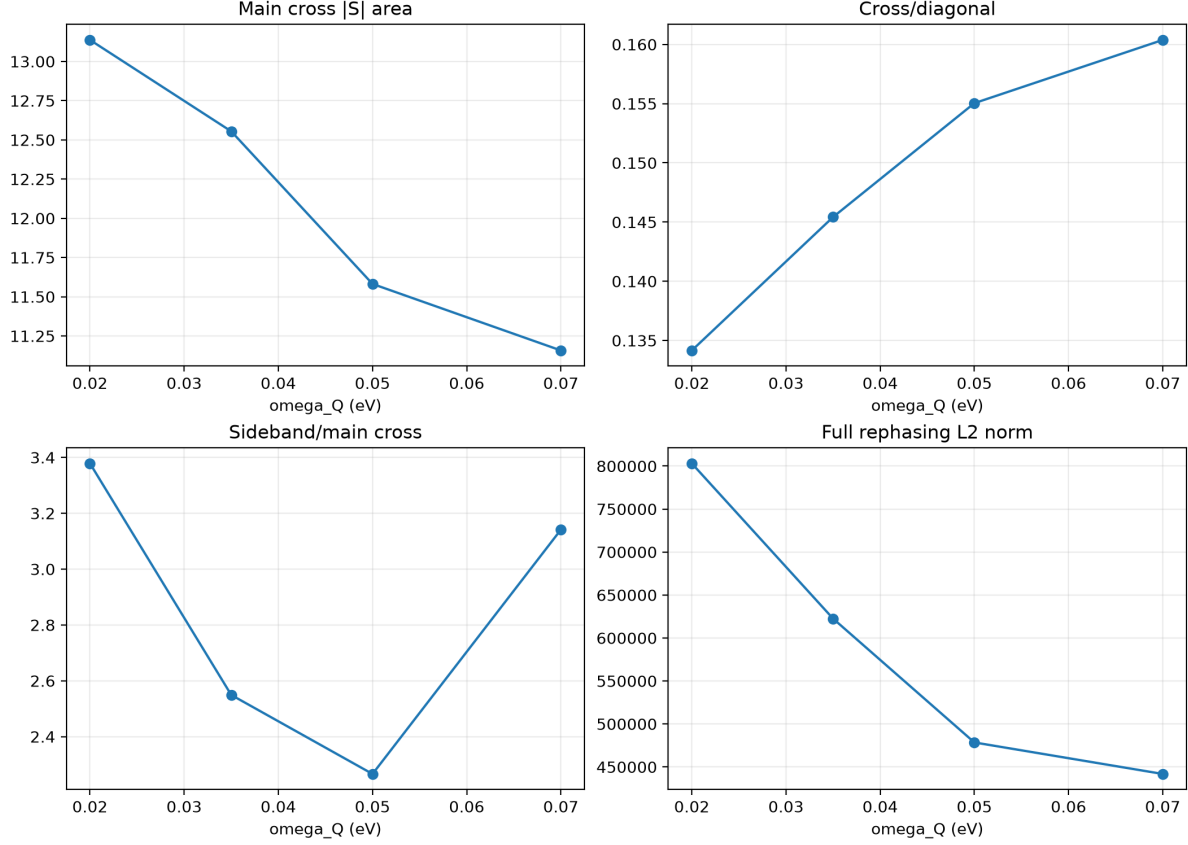


Figure 2: Scan of ω_Q at fixed $V_{BD}^{\text{static}} = 0.02$ and $g_Q = 0.010$.

alone. The most robust indicators in this scan are the full rephasing norm and the cross/diagonal ratio.

The ω_Q criterion also passes at the level of trend sensitivity. The sideband/main-cross ratio varies substantially across the scan, and the main cross-window area, cross/diagonal ratio, and full rephasing norm also change with ω_Q . The sideband metric should not yet be read as a precise phonon-energy measurement because it is based on finite windows rather than a peak-fitting procedure.

Overall, Step 6 confirms that the dynamic phonon coordinate produces spectral information beyond the scalar V_{BD}^{static} channel. The next technical refinement should be a more targeted sideband or peak-tracking analysis, especially if ω_Q is to be extracted quantitatively.